

MLOps Assignment 2

Binary Image Classification (Cats vs Dogs)

Group Details:

Group Number: 3

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1. Project Overview

Objective: Design and implement an end-to-end MLOps pipeline for a binary image classification system (Cats vs Dogs) for a pet adoption platform.

Scope:

- Data & code versioning (Git + DVC)
- Model development and experiment tracking
- FastAPI inference service
- Containerization using Docker
- CI pipeline with GitHub Actions
- CD / deployment workflow
- Monitoring, logging, and performance tracking

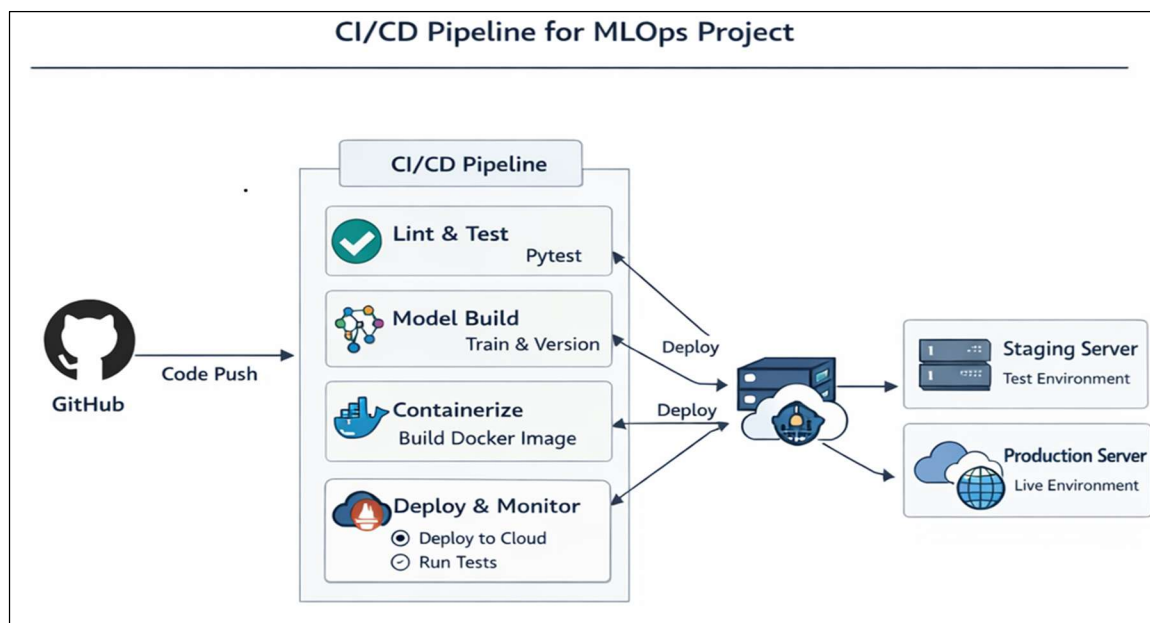
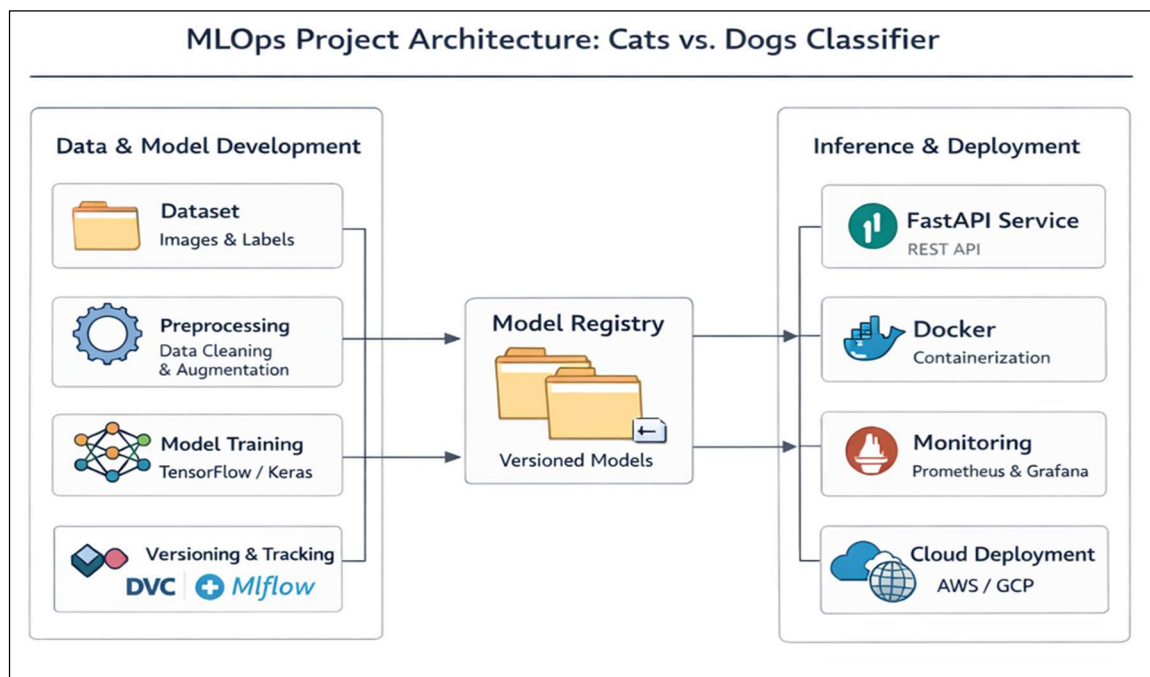
Use Case:

Users upload an image; the system predicts whether the image contains a cat or a dog and returns a confidence score.

2. Dataset :

- Cats vs Dogs dataset (Kaggle)
- Images preprocessed to 224x224 RGB
- Train/Validation/Test split (80/10/10)
- Data augmentation used for generalization

3. High-Level Architecture



Key idea:

- Git tracks code and lightweight metadata (.dvc files)
- DVC tracks large artifacts (datasets, models)
- CI validates code via tests before packaging
- CD deploys containerized inference service

4. Project Structure

binary-image-classification-mlops/

```
├── src/
│   ├── train.py
│   ├── evaluate.py
│   ├── inference.py
│   ├── versioning.py
│   └── performance_tracking.py
├── tests/
├── models/
├── artifacts/
├── configs/
├── .github/workflows/
├── Dockerfile
├── docker-compose.yml
├── requirements.txt
├── dvc.yaml
└── README.md
```

Notes:

- src/ contains core ML and API logic
- tests/ contains unit tests (pytest)
- models/ managed by DVC (NOT Git)
- artifacts/ stores evaluation outputs (plots, reports)

5. M1 — Model Development & Experiment Tracking

Data & Code Versioning:

- Git used for source code and pipeline definitions
- DVC used for dataset and model versioning

Typical commands:

```
dvc add data/
```

```
git add data.dvc .gitignore
```

```
dvc push
```

Model Building:

- Baseline CNN implemented with TensorFlow/Keras
- Input shape: 224x224x3
- Binary output (cat vs dog)
- Model saved as model.h5

Experiment Tracking:

- MLflow used to log runs, parameters, metrics, and artifacts
- Logged artifacts include confusion matrix, ROC curve, and evaluation reports

6. M2 — Model Packaging & Containerization

Inference Service:

- FastAPI application exposes:

```
GET /health
```

```
POST /predict
```

```
GET /performance
```

```
GET /metrics
```

Important design decisions:

- Lazy model loading to avoid CI/test failures
- Safe Prometheus metric registration to avoid duplicate timeseries

Environment Specification:

- requirements.txt with pinned versions for reproducibility

Containerization:

- Dockerfile builds a reproducible inference image
- Local validation:

```
docker build -t catsdogs-api .
```

```
docker run -p 8000:8000 catsdogs-api
```

7. M3 — CI Pipeline (GitHub Actions)

Pipeline triggers:

- Push / PR affecting src/**, tests/**, requirements.txt

CI steps:

1. Checkout repository
2. Setup Python 3.10
3. Install dependencies
4. Run pytest
5. Build Docker image

Testing:

- Unit tests for preprocessing and inference utilities
- API import safety validated by tests

Outcome:

- Failing tests block build and deployment

8. M4 — CD Pipeline & Deployment

Deployment target:

- Docker Compose or Kubernetes (as configured)

CD flow:

- New image built in CI
- Image pushed to registry
- Deployment updates running service

Smoke tests:

- Health endpoint call
- One prediction call
- Pipeline fails if smoke tests fail

9. M5 — Monitoring, Logging & Performance Tracking

Logging:

- Request-level logging with latency and request count

Prometheus Metrics:

- prediction_requests_total
- prediction_latency_seconds

Post-deployment performance tracking:

- true label + predicted label logging
- summary endpoint for monitoring model quality

10. Key Issues Faced & Fixes

Issue 1: GitHub large file limit (>100MB)

Fix: Removed model from Git history; tracked via DVC.

Issue 2: Secret scanning blocked push

Fix: Removed service account JSON from history; added to .gitignore.

Issue 3: CI failures due to model loading at import time

Fix: Lazy model loading and safe checks for missing model.

Issue 4: Prometheus duplicated timeseries during pytest

Fix: Safe metric creation using REGISTRY fallback.

11. How to Run the Project Locally

1. Clone repository

```
git clone <repo-url>
```

2. Create virtual environment

```
python -m venv .venv
```

```
source .venv/bin/activate (Linux/Mac)
```

```
.venv\Scripts\activate (Windows)
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. Pull data/models (if using DVC remote)

```
dvc pull
```

5. Run API

```
uvicorn src.inference:app --reload
```

6. Run tests

```
pytest -v
```

12. API Usage Examples

Health check:

GET /health

Prediction:

POST /predict

Form-data:

file=<image>

Example response:

```
{  
  "label": "dog",  
  "confidence": 0.94,  
  "model_version": "v1"  
}
```

Metrics:

GET /metrics

13. Future Improvements

- Model registry integration (MLflow Model Registry)
- Automated retraining pipeline
- Data drift detection
- Canary deployment strategy
- Kubernetes autoscaling
- Caching DVC artifacts in CI for faster builds

14. Conclusion

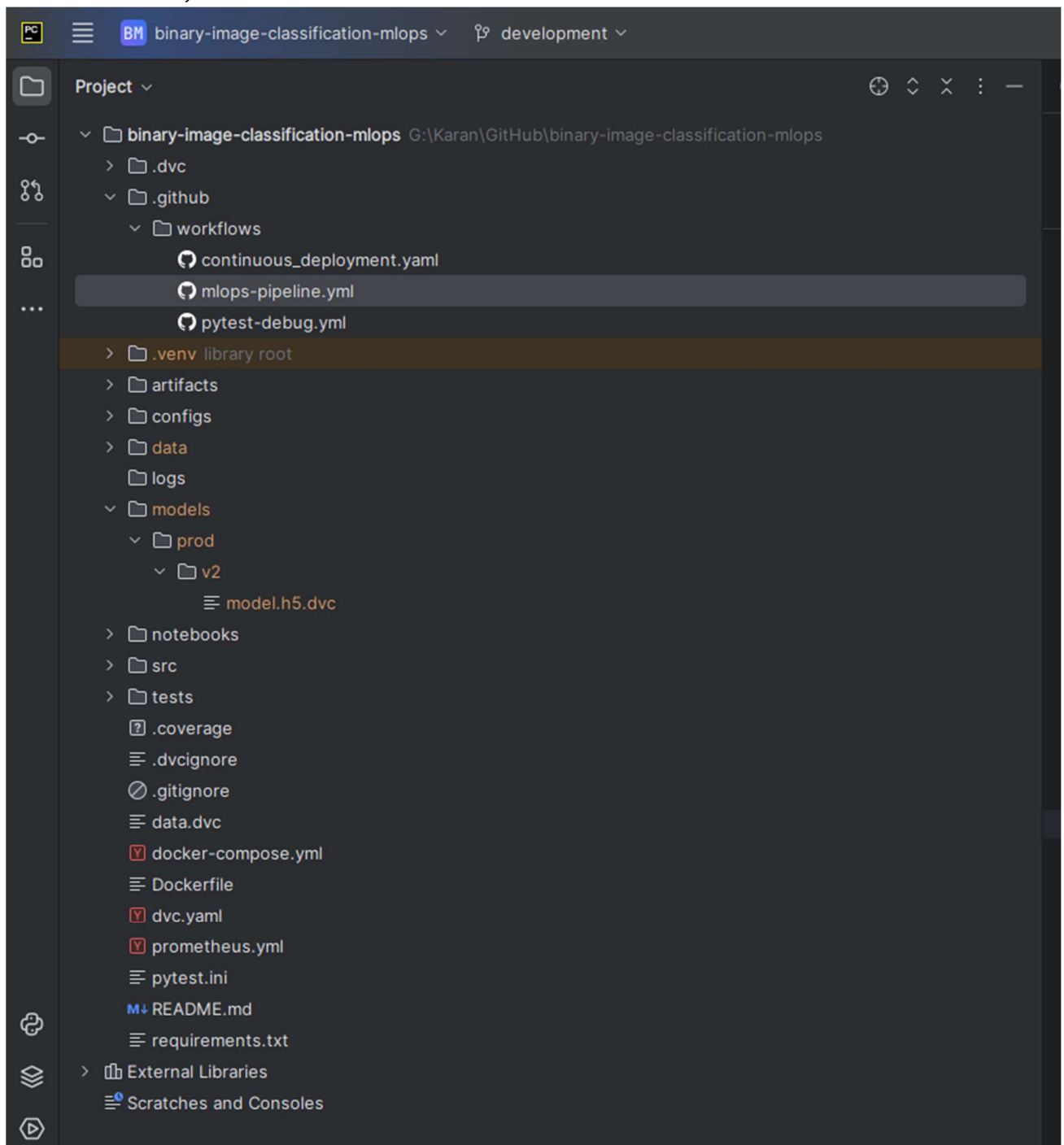
This project demonstrates a complete MLOps lifecycle:

data versioning → training → experiment tracking → API packaging → containerization → CI/CD → monitoring.

The system is designed to be reproducible, testable, and deployable with modern open-source MLOps tools.

15. Screenshots

1. Screenshot 1: Project folder structure



2. Screenshot 2: DVC setup and dvc status

```
Project
Terminal Local x + v
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops>
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> dvc status
Data and pipelines are up to date.
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> dvc remote list
storage gdrive://1YtmChnMxhDgP26Bf0bRpR3LMK0Ku0Itn
myremote s3://binary-image-mlops-dvc (default)
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> dvc config --list
remote.storage.url=gdrive://1YtmChnMxhDgP26Bf0bRpR3LMK0Ku0Itn
remote.storage.gdrive_use_service_account=false
remote.myremote.url=s3://binary-image-mlops-dvc
remote.myremote.region=ap-south-1
core.remote=myremote
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> dvc status
Data and pipelines are up to date.
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> dvc list . data
processed
raw
(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> ls *.dvc

Directory: G:\Karan\GitHub\binary-image-classification-mlops

Mode                LastWriteTime         Length Name
----                -
d-----         21-02-2026      18:33             .dvc
-a-----         21-02-2026      17:40          115 data.dvc

(.venv) PS G:\Karan\GitHub\binary-image-classification-mlops> 
```

3. Screenshot 3: MLflow experiment tracking dashboard

127.0.0.1:5000/#/experiments/213038925468938809?searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFilter=Active&datasetsFilter=W10%3D&modelVersionFilter=All%20Runs&selectedColumn=...

mlflow 2.9.2 Experiments Models

Experiments

cats_dogs_dev

cats_dogs_dev Provide Feedback

Experiment ID: 213038925468938809 Artifact Location: file:///G:/Karan/GitHub/binary-image-classification-mlops/mlruns/213038925468938809

Description Edit

metrics.rmse < 1 and params.model = "tree" Time created State: Active Sort: Created Columns

Table Chart Evaluation Experimental

	Run Name	Created	Dataset	Duration	Source	Models
	dev_v2_evaluation	3 minutes ago	-	2.6s	G:\Karan...	-
	dev_v2	10 minutes ago	-	6.0min	G:\Karan...	tensorflow
	dev_v1	21 minutes ago	-		G:\Karan...	-

127.0.0.1:5000/#/experiments/213038925468938809/runs/84421bbb83914a82aea95e735d7c507c

mlflow 2.9.2 Experiments Models

cats_dogs_dev >

dev_v2_evaluation

Run ID: 84421bbb83914a82aea95e735d7c507c Date: 2026-02-21 22:31:41 Source: G:\Karan\GitHub\binary-image-classification-mlops\src\evaluate.py

Duration: 2.6s Status: FINISHED Lifecycle Stage: active

- > Description [Edit](#)
- > Datasets
- > Parameters (1)
- ▼ Metrics (5)

Name	Value
test_accuracy	0.69
test_f1	0.7047619047619048
test_precision	0.6727272727272727
test_recall	0.74
test_roc_auc	0.7316

4. Screenshot 4: Model training logs / metrics

```
WARNING:tensorflow:From G:\Karan\GitHub\binary-image-classification-mlops\.venv\lib\site-packages\keras\src\engine\base_layer_utils.py:38:
xecuting_eagerly_outside_functions instead.

25/25 [=====] - 46s 2s/step - loss: 1.2472 - accuracy: 0.5213 - val_loss: 0.7022 - val_accuracy: 0.5100
Epoch 2/20
25/25 [=====] - 37s 1s/step - loss: 0.6951 - accuracy: 0.5489 - val_loss: 0.6864 - val_accuracy: 0.6100
Epoch 3/20
25/25 [=====] - 37s 1s/step - loss: 0.6737 - accuracy: 0.6053 - val_loss: 0.6386 - val_accuracy: 0.7000
Epoch 4/20
25/25 [=====] - 34s 1s/step - loss: 0.6625 - accuracy: 0.6178 - val_loss: 0.6450 - val_accuracy: 0.6500
Epoch 5/20
25/25 [=====] - 34s 1s/step - loss: 0.6448 - accuracy: 0.6441 - val_loss: 0.6378 - val_accuracy: 0.7000
Epoch 6/20
25/25 [=====] - 36s 1s/step - loss: 0.6283 - accuracy: 0.6404 - val_loss: 0.6121 - val_accuracy: 0.6900
Epoch 7/20
25/25 [=====] - 38s 2s/step - loss: 0.6080 - accuracy: 0.6742 - val_loss: 0.6285 - val_accuracy: 0.7100
Epoch 8/20
25/25 [=====] - 41s 2s/step - loss: 0.5861 - accuracy: 0.6830 - val_loss: 0.6127 - val_accuracy: 0.6900
Epoch 9/20
25/25 [=====] - 35s 1s/step - loss: 0.5863 - accuracy: 0.6892 - val_loss: 0.6320 - val_accuracy: 0.6900
G:\Karan\GitHub\binary-image-classification-mlops\.venv\lib\site-packages\keras\src\engine\training.py:3103: UserWarning: You are saving
command using instead the native Keras format, e.g. 'model.save('my_model.keras')'.
saving_api.save_model(
```

```
=====
✅Evaluation Completed
Environment : dev
Version    : v2
Accuracy   : 0.6900
ROC-AUC    : 0.7316
Promote    : True
=====
```

	precision	recall	f1-score	support
Cat	0.71	0.64	0.67	50
Dog	0.67	0.74	0.70	50
accuracy			0.69	100
macro avg	0.69	0.69	0.69	100
weighted avg	0.69	0.69	0.69	100

▼ Artifacts

▼ evaluation

- 📄 classification_report_dev_v2.txt
- 📄 confusion_matrix_dev_v2.png
- 📄 roc_curve_dev_v2.png

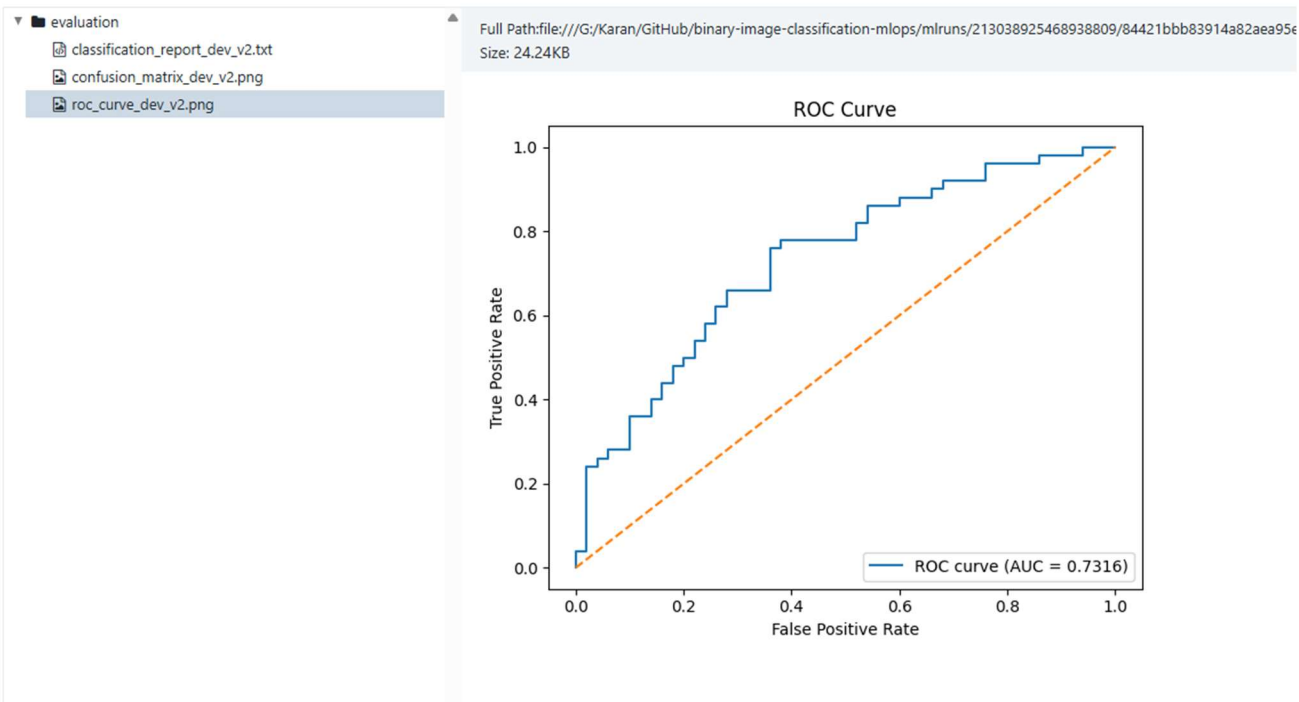
Full Path:file:///G:/Karan/GitHub/binary-image-classification-mlops/mlruns/213038
Size: 334B

	precision	recall	f1-score	support
Cat	0.71	0.64	0.67	50
Dog	0.67	0.74	0.70	50
accuracy			0.69	100
macro avg	0.69	0.69	0.69	100
weighted avg	0.69	0.69	0.69	100

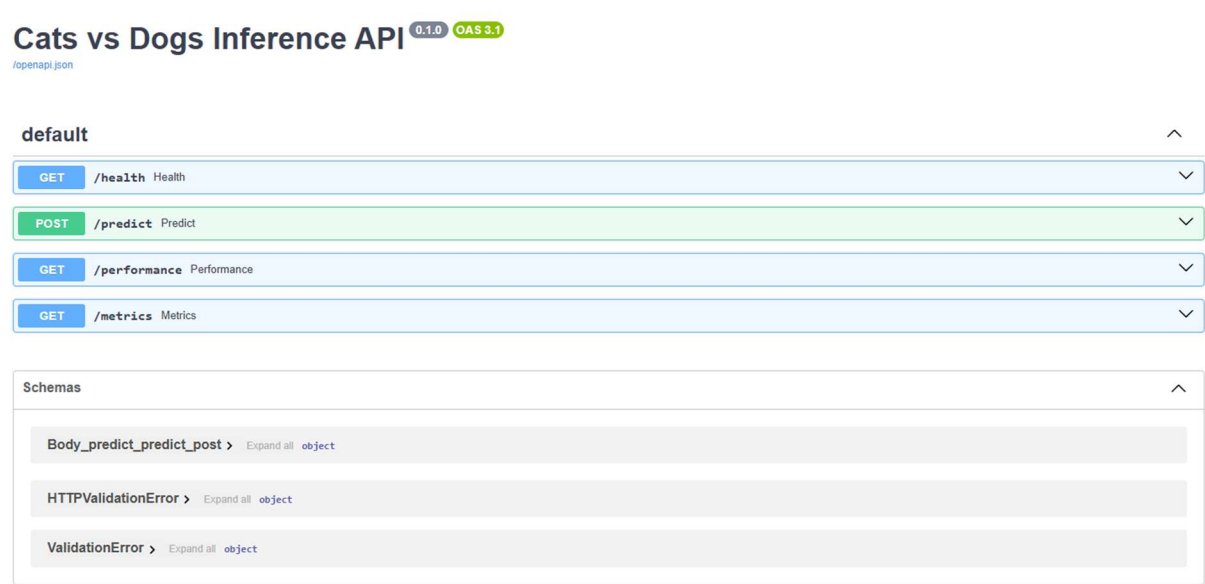
▼ Artifacts



▼ Artifacts



5. Screenshot 5: FastAPI /docs page



6. Screenshot 8: Docker build success



7. Screenshot 9: Docker container running


```
Run tests with coverage

14 platform linux -- Python 3.10.19, pytest-7.4.2, pluggy-1.6.0
15 rootdir: /home/runner/work/binary-image-classification-mlops/binary-image-classification-mlops
16 configfile: pytest.ini
17 plugins: hydra-core-1.3.2, anyio-3.7.1, cov-4.1.0
18 collected 12 items
19
20 tests/test_api.py .. [ 16%]
21 tests/test_evaluate.py . [ 25%]
22 tests/test_inference.py . [ 33%]
23 tests/test_model_performance.py . [ 41%]
24 tests/test_preprocessing.py ... [ 66%]
25 tests/test_promote.py . [ 75%]
26 tests/test_train.py . [ 83%]
27 tests/test_versioning.py .. [100%]
28
29 ===== warnings summary =====
30 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/fastapi/openapi/models.py:55
31 /opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/fastapi/openapi/models.py:55: DeprecationWarning: `general_plain_validator_function` is deprecated, use
32     return general_plain_validator_function(cls._validate)
33
34 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/pydantic_core/core_schema.py:4018
35 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/pydantic_core/core_schema.py:4018
36 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/pydantic_core/core_schema.py:4018
37 /opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/pydantic_core/core_schema.py:4018: DeprecationWarning: `general_plain_validator_function` is deprecated, use
38     warnings.warn(
39
40 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/fastapi/datastructures.py:52
41 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/fastapi/datastructures.py:52
42 /opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/fastapi/datastructures.py:52: DeprecationWarning: `general_plain_validator_function` is deprecated, use
43     return general_plain_validator_function(cls._validate)
44
45 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/httpx/_client.py:680
46 /opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/httpx/_client.py:680: DeprecationWarning: The 'app' shortcut is now deprecated. Use the explicit style
47     warnings.warn(message, DeprecationWarning)
48
49 ../../../../opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/mlflow/utils/requirements_utils.py:20
50 /opt/hostedtoolcache/Python/3.10.19/x64/lib/python3.10/site-packages/mlflow/utils/requirements_utils.py:20: DeprecationWarning: pkg_resources is deprecated as an API. See
51     import pkg_resources # noqa: TID251
52
53 ..
```

10. Screenshot 12: Docker image publish step (if applicable)

```
Push Docker image

1 ▶ Run docker push ***/binary-image-classification-mlops:binary-image-classification-mlops-development-v1
12 The push refers to repository [docker.io/***/binary-image-classification-mlops]
13 7f7e9b0257c5: Preparing
14 01a2ac1a8440: Preparing
15 3b2b23c87b75: Preparing
16 3245017f8af7: Preparing
17 766ae7f1b0fe: Preparing
18 e95094b3ff52: Preparing
19 080d20b5229f: Preparing
20 bb2cb0bd0677: Preparing
21 9f0194fcc39b: Preparing
22 2cb59db770d1: Preparing
23 a8ff6f8cbdfd: Preparing
24 e95094b3ff52: waiting
25 080d20b5229f: waiting
26 2cb59db770d1: waiting
27 a8ff6f8cbdfd: waiting
28 bb2cb0bd0677: waiting
29 9f0194fcc39b: waiting
30 3b2b23c87b75: Pushed
31 766ae7f1b0fe: Pushed
32 7f7e9b0257c5: Pushed
33 bb2cb0bd0677: Mounted from library/python
34 9f0194fcc39b: Mounted from library/python
35 2cb59db770d1: Mounted from library/python
36 a8ff6f8cbdfd: Mounted from library/python
37 080d20b5229f: Pushed
38 e95094b3ff52: Pushed
39 01a2ac1a8440: Pushed
40 3245017f8af7: Pushed
41 binary-image-classification-mlops-development-v1: digest: sha256:673ad6029e0bf68dc236f41f12dfc9b86082e0e2198bd6c4b4bb1659b7228474 size: 2627
```

11. Screenshot 13: Deployment running (Docker Compose/K8s)

Cats vs Dogs Inference API 0.1.0 OAS 3.1

/openapi.json

default

GET	/health	Health	
POST	/predict	Predict	
GET	/performance	Performance	
GET	/metrics	Metrics	

Schemas

Body_predict_predict_post > Expand all object

HTTPValidationError > Expand all object

ValidationError > Expand all object

127.0.0.1:8000/docs#/default/health_health_get

GET /health Health

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/health' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/health
```

Server response

Code Details

200

Response body

```
{
  "status": "healthy",
  "model_version": "v2"
}
```

Response headers

```
content-length: 41
content-type: application/json
date: Sat, 21 Feb 2026 17:45:25 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@17.jpg;type=Image/jpeg'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code	Details
------	---------

200	Response body
-----	---------------

```
{
  "label": "dog",
  "confidence": 0.6325368285179139,
  "model_version": "v2",
  "latency": 0.8667945861816406,
  "request_count": 1
}
```

Response headers

```
content-length: 115
content-type: application/json
date: Sat, 21 Feb 2020 17:55:27 GMT
server: uvicorn
```

Responses

Code	Description	Links
------	-------------	-------

200	Successful Response	No links
-----	---------------------	----------

Media type

application/json

Controls Accept header.

Example Value | Schema

Curl

```
curl -X 'GET' \
  'http://localhost:8000/metrics' \
  -H 'accept: application/json'
```

Request URL

http://localhost:8000/metrics

Server response

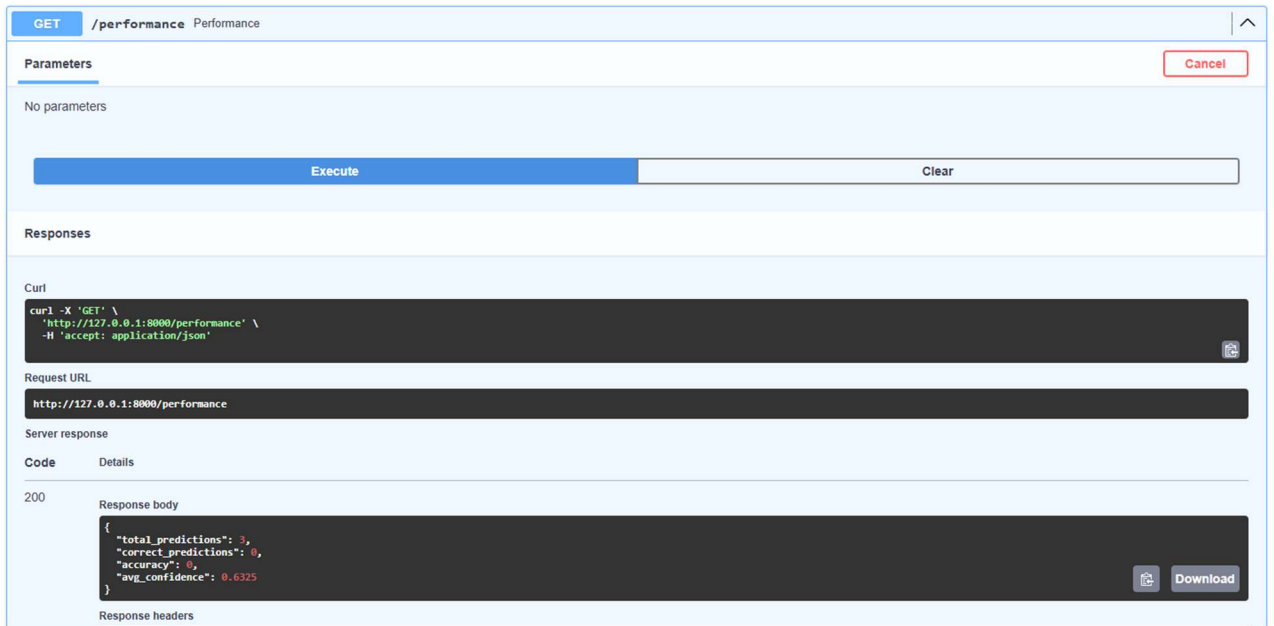
Code	Details
------	---------

200	Response body
-----	---------------

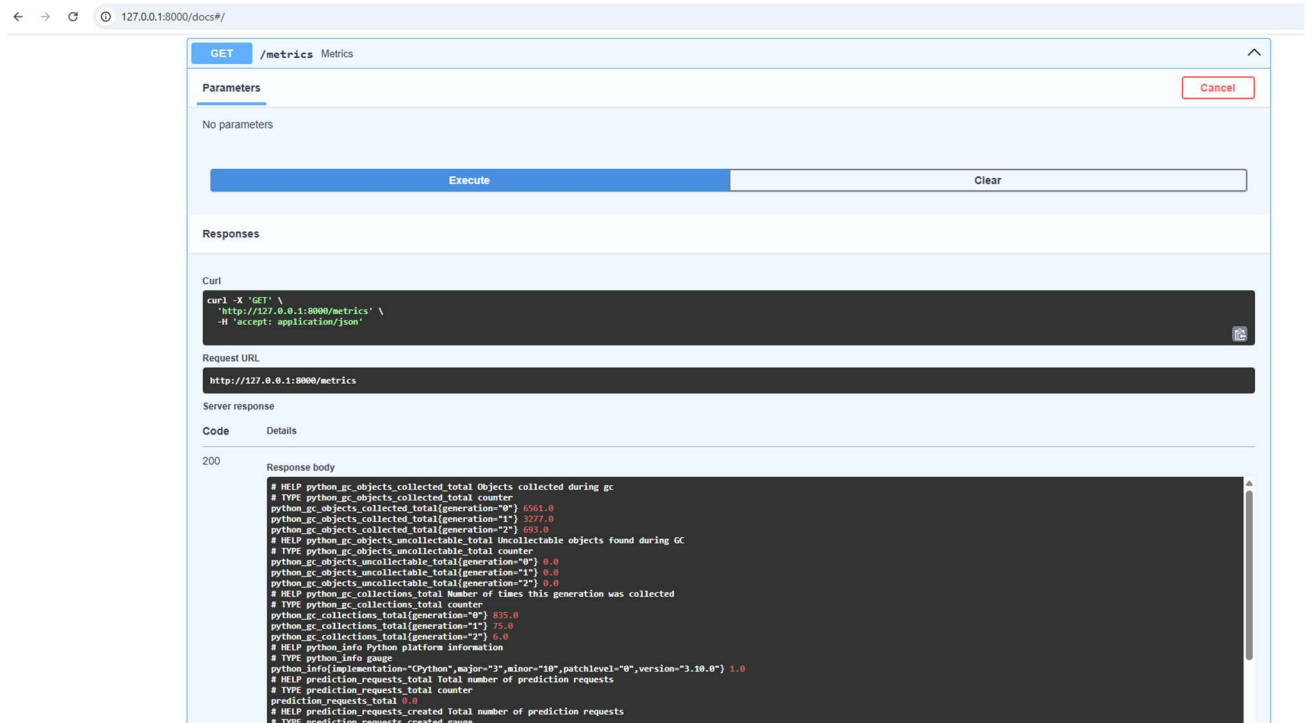
```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 2656.0
python_gc_objects_collected_total{generation="1"} 3344.0
python_gc_objects_collected_total{generation="2"} 330.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 779.0
python_gc_collections_total{generation="1"} 70.0
python_gc_collections_total{generation="2"} 5.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="10",patchlevel="19",version="3.10.19"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 2.105479168e+09
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 5.53058304e+08
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
```

Response headers

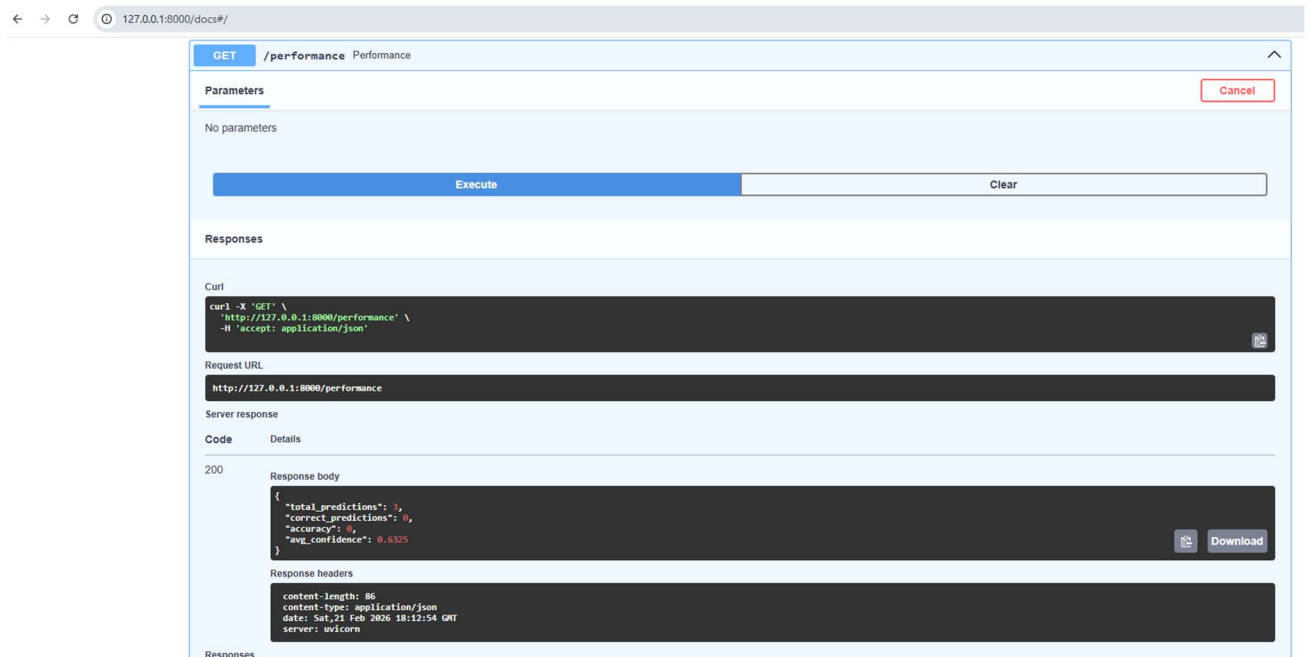
```
content-length: 3316
content-type: text/plain; version=0.0.4; charset=utf-8
```



12. Screenshot 15: Prometheus metrics endpoint



13. Screenshot 16: Performance summary endpoint



16. Repository, Docker & Video Links

Git Hub Repository: <https://github.com/KSharma-SourceCode/binary-image-classification-mlops>

Docker Hub Image Registry:

<https://hub.docker.com/repository/docker/ksharmadockerhub/binary-image-classification-mlops/general>

CI/CD and Deployment Workflow: <https://github.com/KSharma-SourceCode/binary-image-classification-mlops/actions>

Video Link:

Google Drive:

<https://drive.google.com/file/d/18WJlB-jpkpWz8dW5ikWm6f63RmHeY5fu/view?usp=sharing>